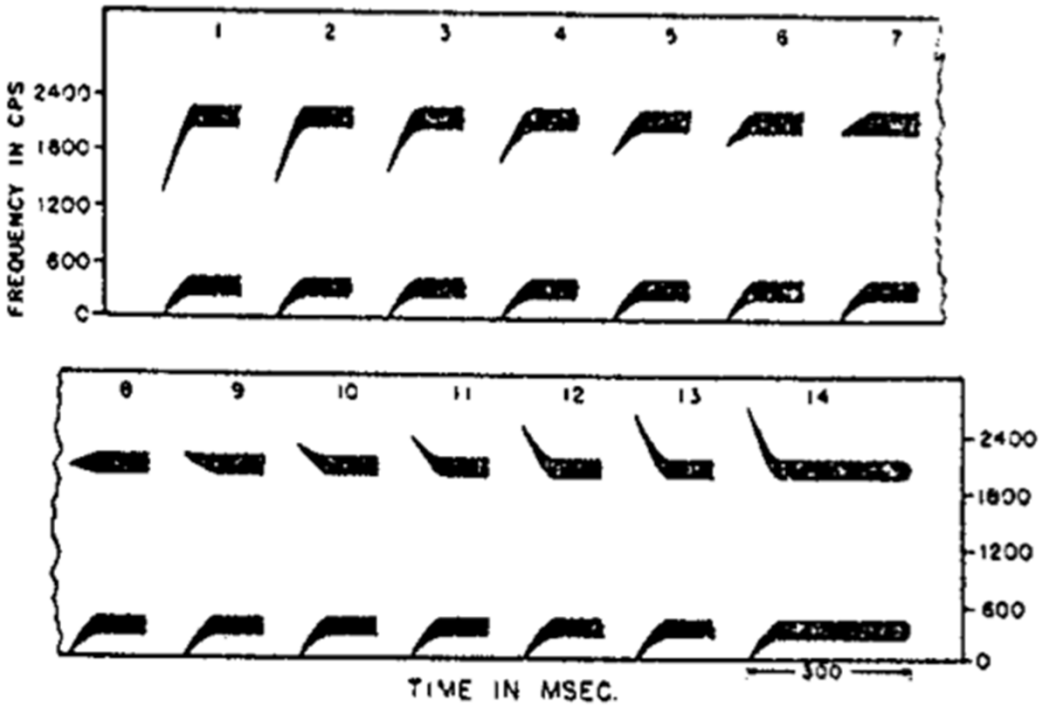
Psyc/ling 406 Midterm 2 review

Fall 2025 (20253)

Categorical perception

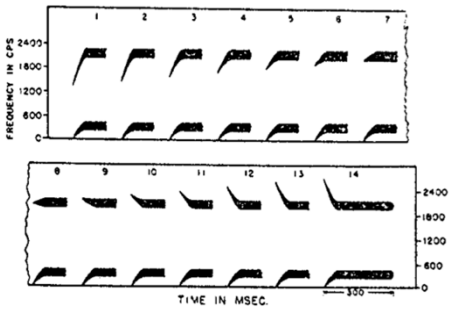


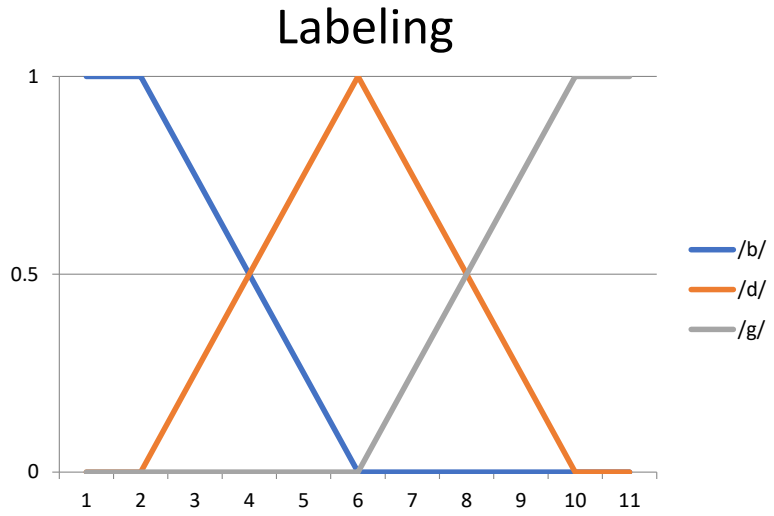
A speech ***continuum:***

These stimuli are constructed by keeping F1 constant and varying F2 onset frequency *linearly* from stimulus 1 to 13.

Thus: adjacent stimuli differ by *the same amount* across the entire continuum.

A simple “ideal observer” model assuming perception is *non-categorical*.

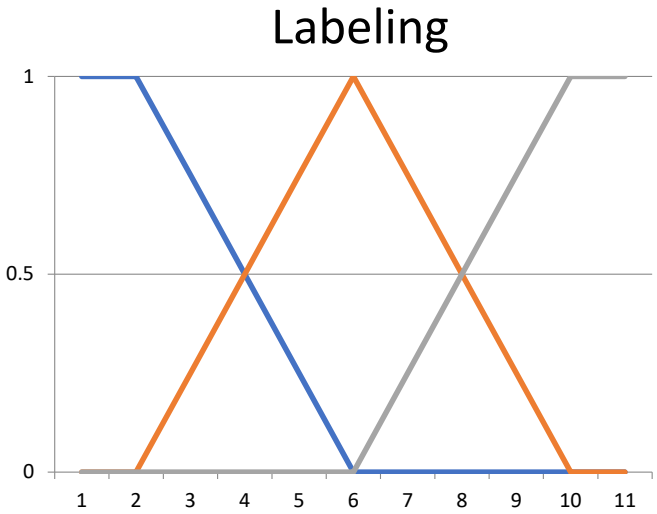


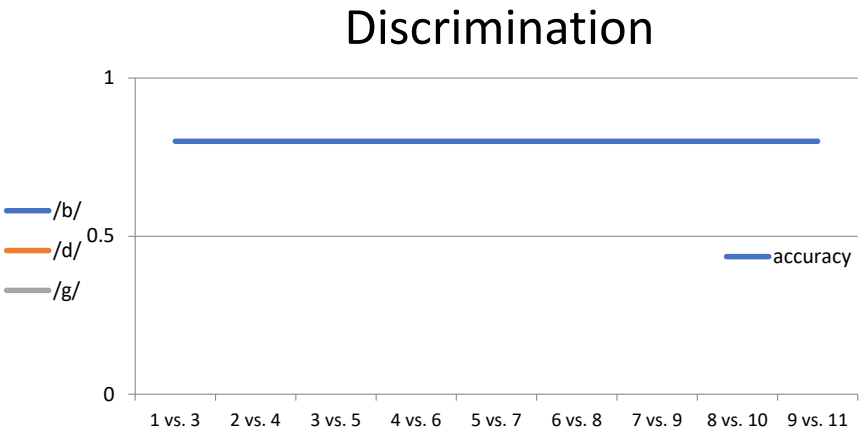


The first thing we’d predict is that categorization would get “blurry” as you move away from category prototypes. We know the endpoints are /b/ and /g/, with /d/ in the middle. Between these points, stimulus ambiguity should scale with the distance from the prototype. So, stimulus 2 would be 75/25 /b/ - /d/.

Liberman et al. (1957) available here: http://www.haskins.yale.edu/Reprints/HL0022.pdf

A simple “ideal observer” model assuming perception is linear.

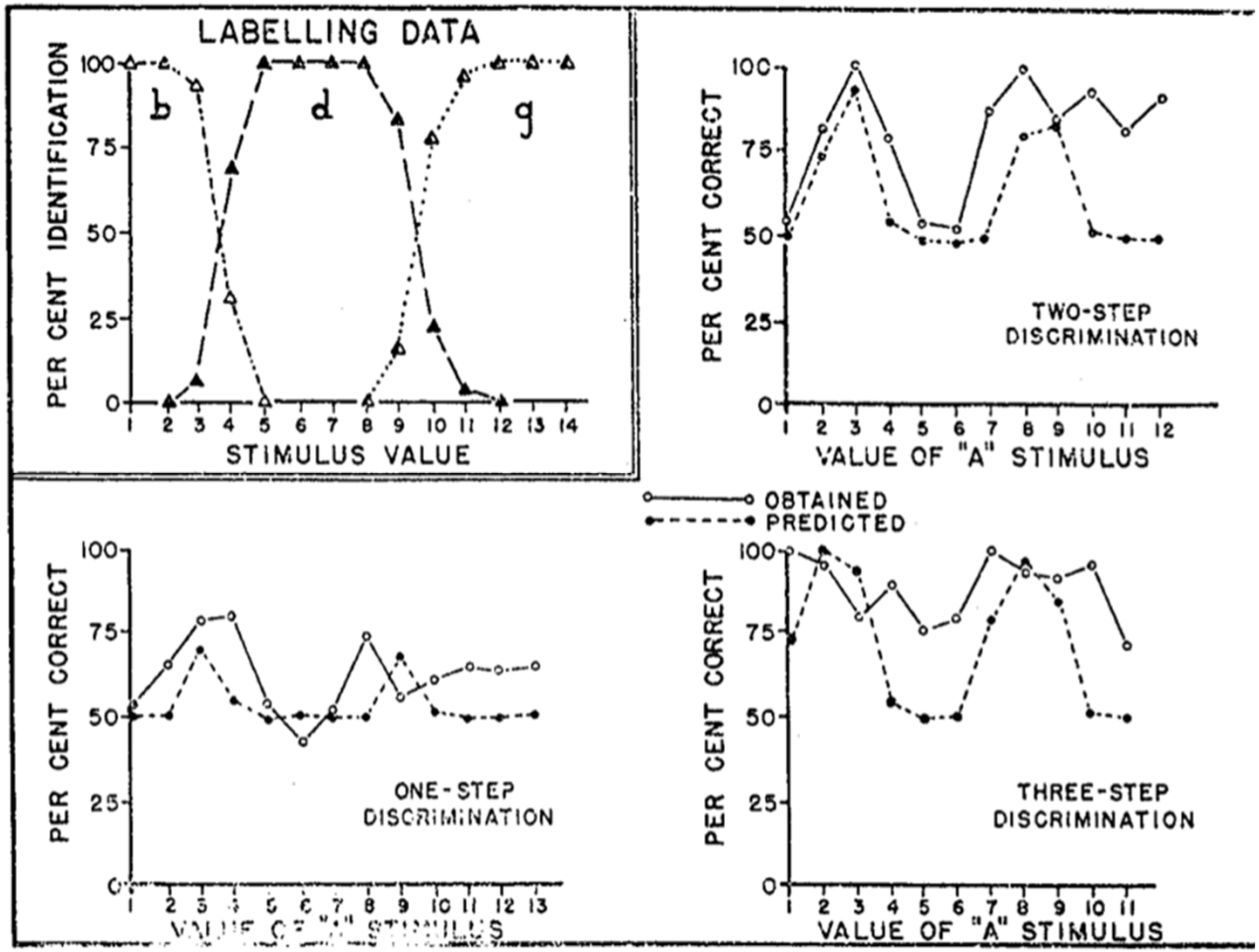




But the other important thing we’d predict is that Discrimination performance should be independent of Labeling.

Obviously labeling has to have a sort of “peaked” function, because you’ve got three categories, and you’re asking people to make binary responses. But for the discrimination task, you’re just asking people to say “same” or “different.” If perception of differences in F2 onset were constant over the whole continuum, discrimination accuracy would also be constant.

Categorical perception



Liberman et al. (1957) available here: http://www.haskins.yale.edu/Reprints/HL0022.pdf

That's one piece of evidence for *cascaded* processing in speech perception.

It would be hard to explain why the number of looks to the other member of the minimal pair increases gradually as you approach the category boundary in a thresholded model…

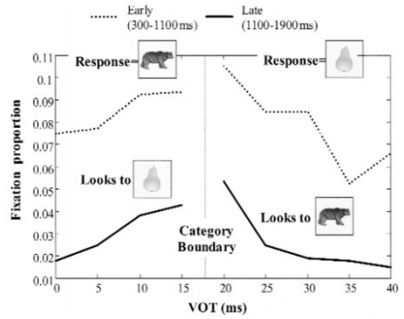
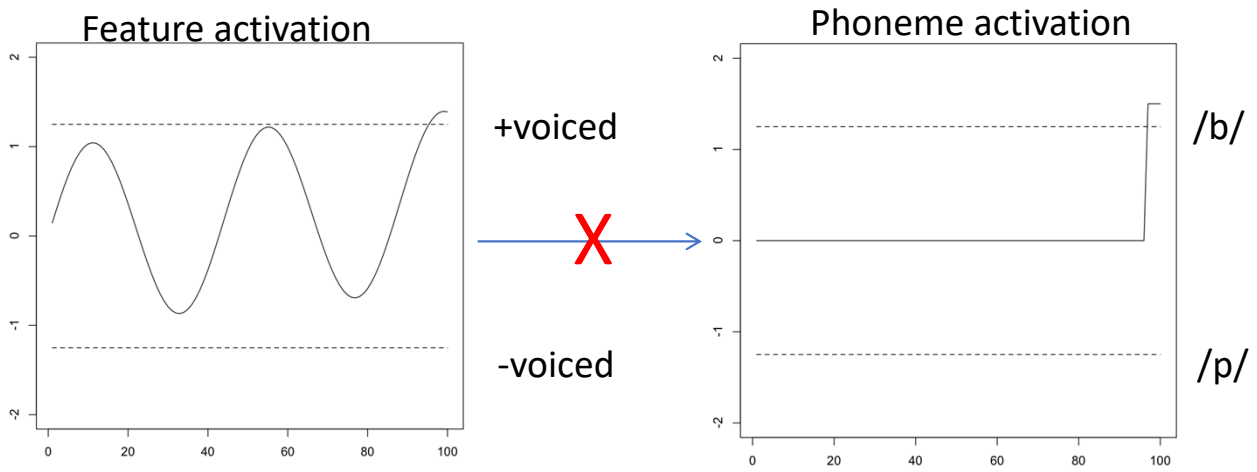
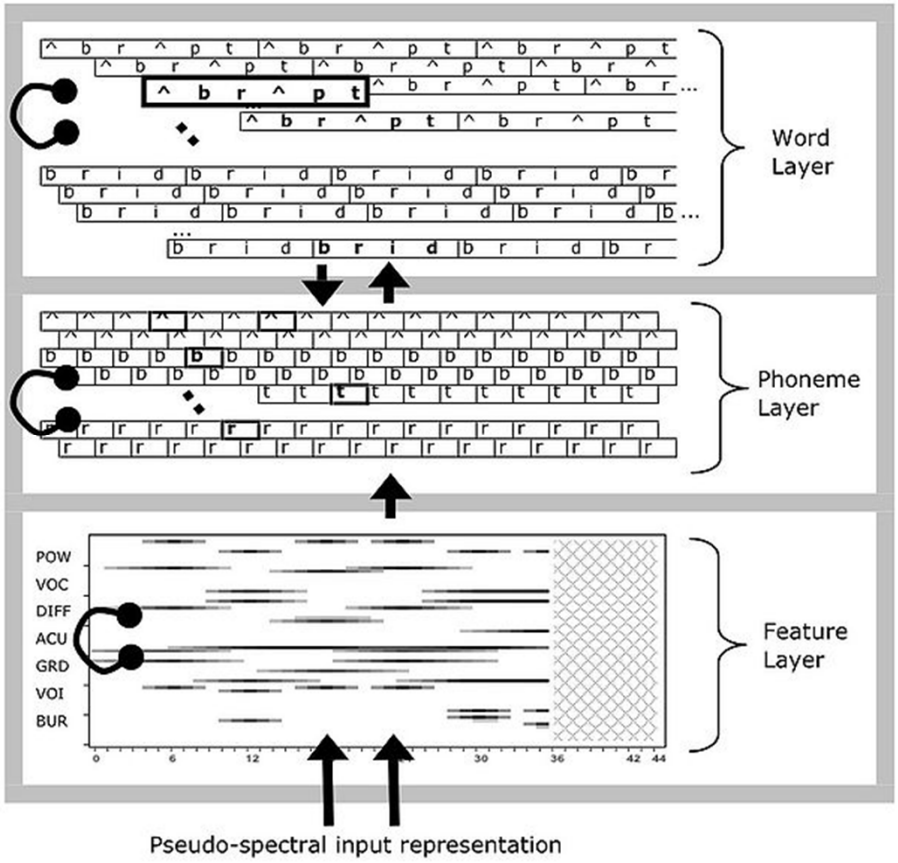


Fig. 6. Effect of VOT and time on fixations to the competitor.

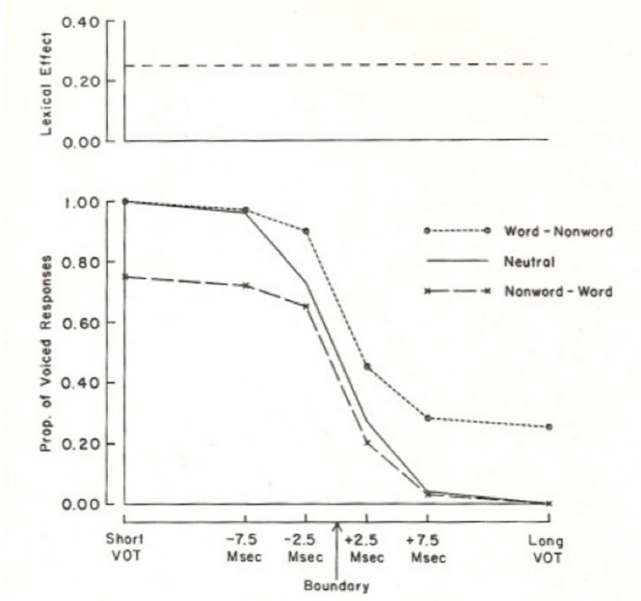


Note:
This model also incorporates *feedback* from the lexical layer to the phoneme layer



Next, we'll take a look at some experiments that motivate incorporating this kind of feedback.

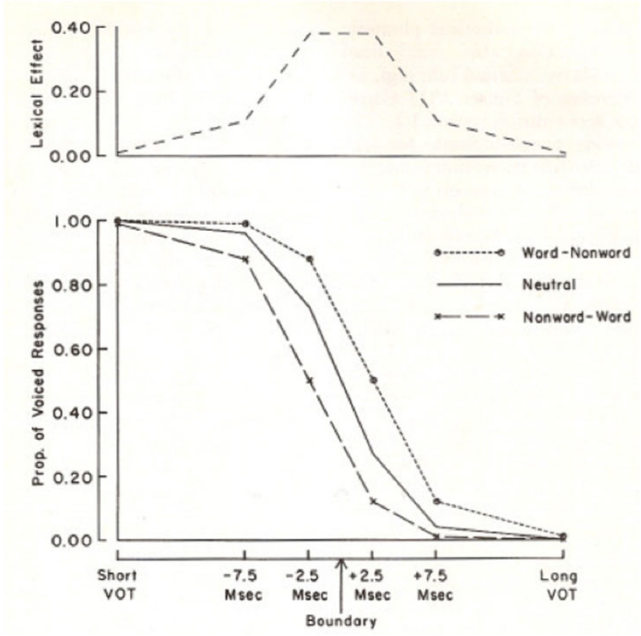
Gisses and kifts for Valentine's day: The Ganong effect.



Ganong (1980) looked at how phoneme identification is impacted by word knowledge.

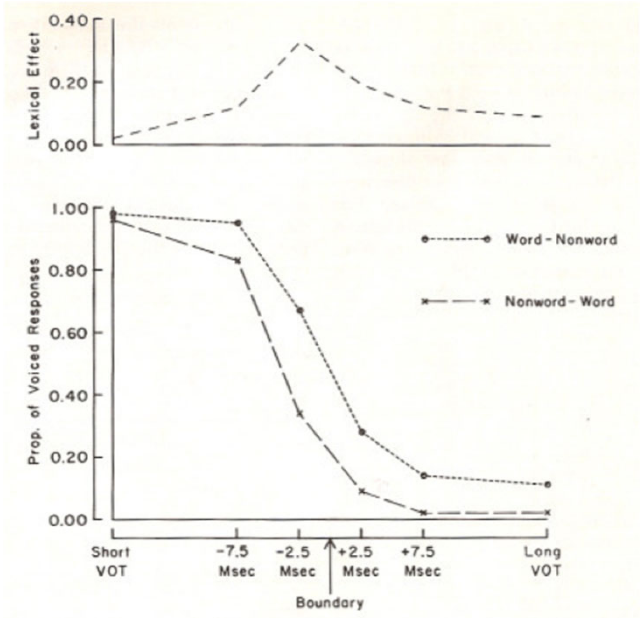
He came up with word/nonword pairs like "corpse/gorpse" and produced continua spanning the relevant phoneme boundary in 5ms steps.

Here's a prediction based on a late, "correction" process, whereby categorical perception functions as usual, but there is a bias to report having heard a word.



Here's a prediction of what would happen if word knowledge instead worked to *shift the categorical boundary.*

Note that the lexical effect is strongest for the items closest to the category boundary, because those are the stimuli for which lexical information can help resolve some ambiguity about the initial phoneme.



The results are pretty strikingly consistent with the second model.

Ganong raises a number of possibilities for how this could happen, focusing on different kinds of guessing strategies.

But later models would emphasize the potential role of feedback in these shifts.

A simulation of the Ganong effect in the TRACE model:

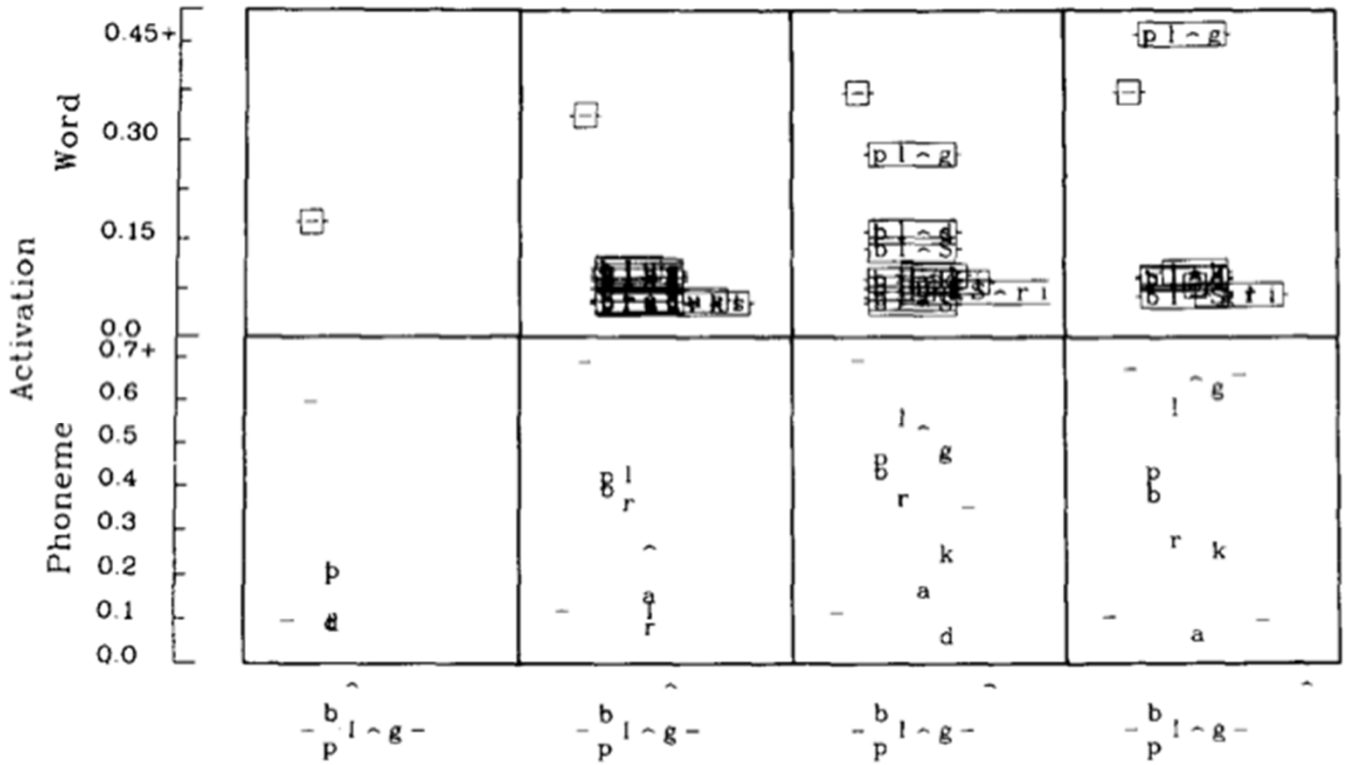


FIG. 5. Phoneme- and word-level activations at several points in the unfolding of a segment ambiguous between /b/ and /p/, followed by /l/, /ˆ/, and /g/. See text for a full explanation.

A simulation of the Ganong effect in the TRACE model:

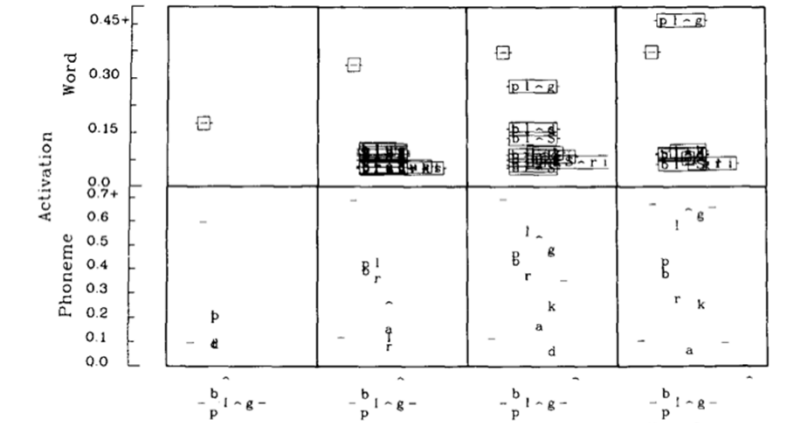
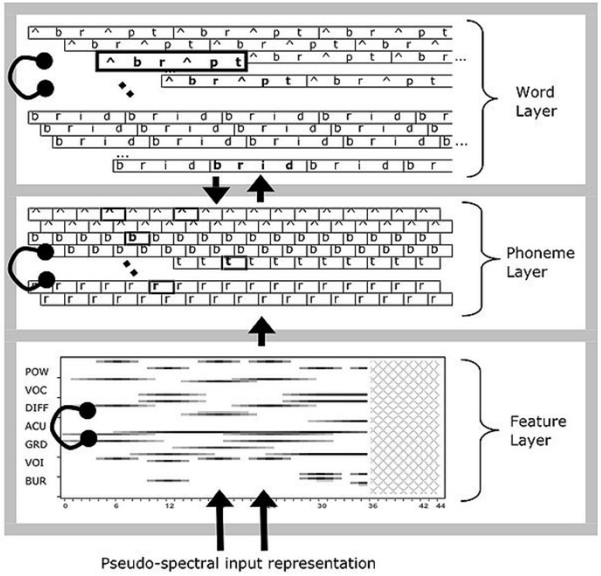


FIG. 5. Phoneme- and word-level activations at several points in the unfolding of a segment ambiguous between /b/ and /p/, followed by /l/, /ɹ/, and /g/. See text for a full explanation.



As the input unfolds over time, activity accumulates for “plug,” especially when the final /g/ arrives, which is highly inconsistent with “blood” and “blush” (also activated in Frame 3).

Thus, this model explains word effects on phoneme category boundaries in terms of *top-down* processing. Specifically, activity from the lexical level can produces *feedback* to the phoneme level, producing greater activation for the phoneme consistent with a word.

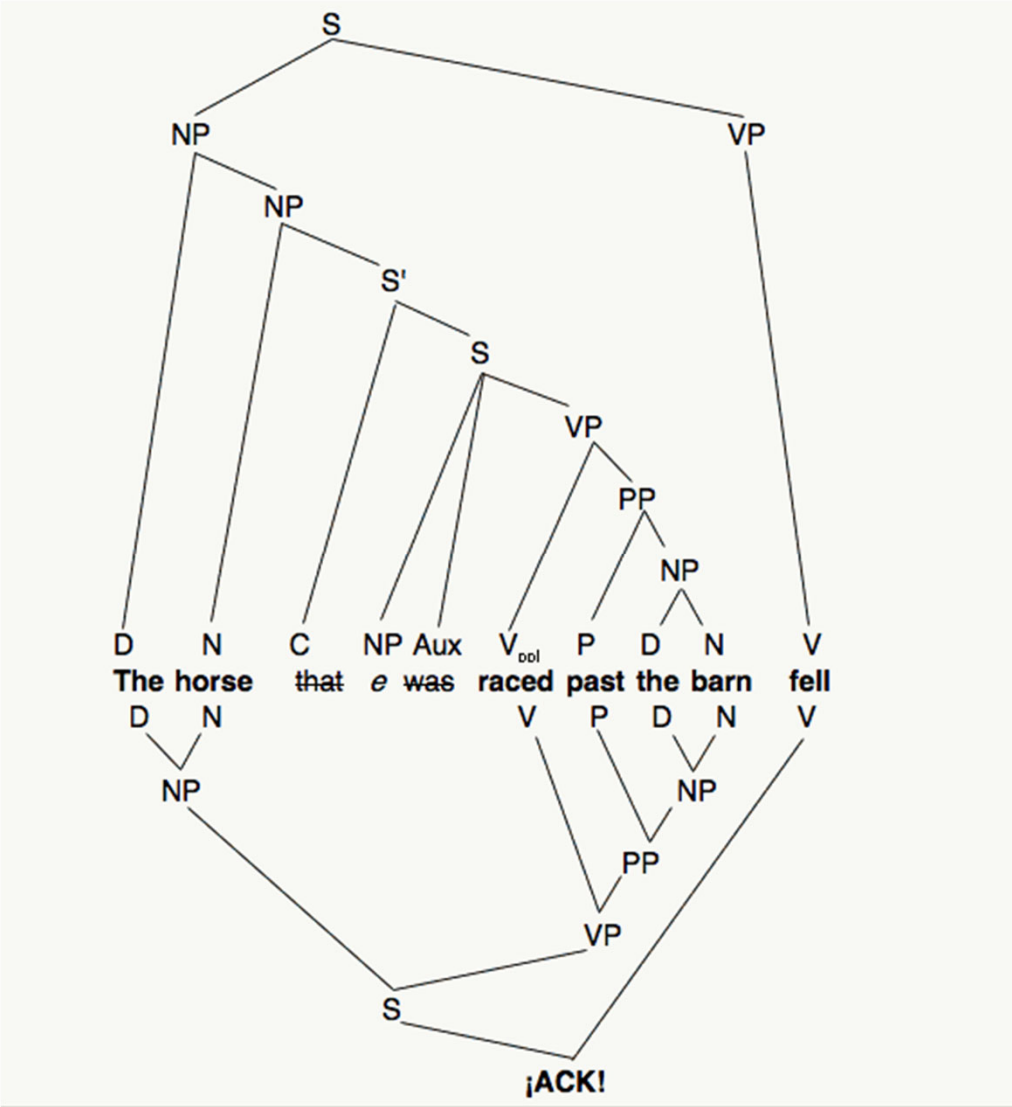
Syntactic explanations of the “garden path” effect

If we imagine that there is a parser whose job is to construct a syntactic tree for the sentences we hear/read, we have to have some way of dealing with the fact that it is implemented in a cognitive system with:

- limited memory
- time constraints

(these trees, and the dinosaur comic are both from the Laguage Log piece on this:

http://languagelog.ldc.upenn.edu/nll/?p=167)



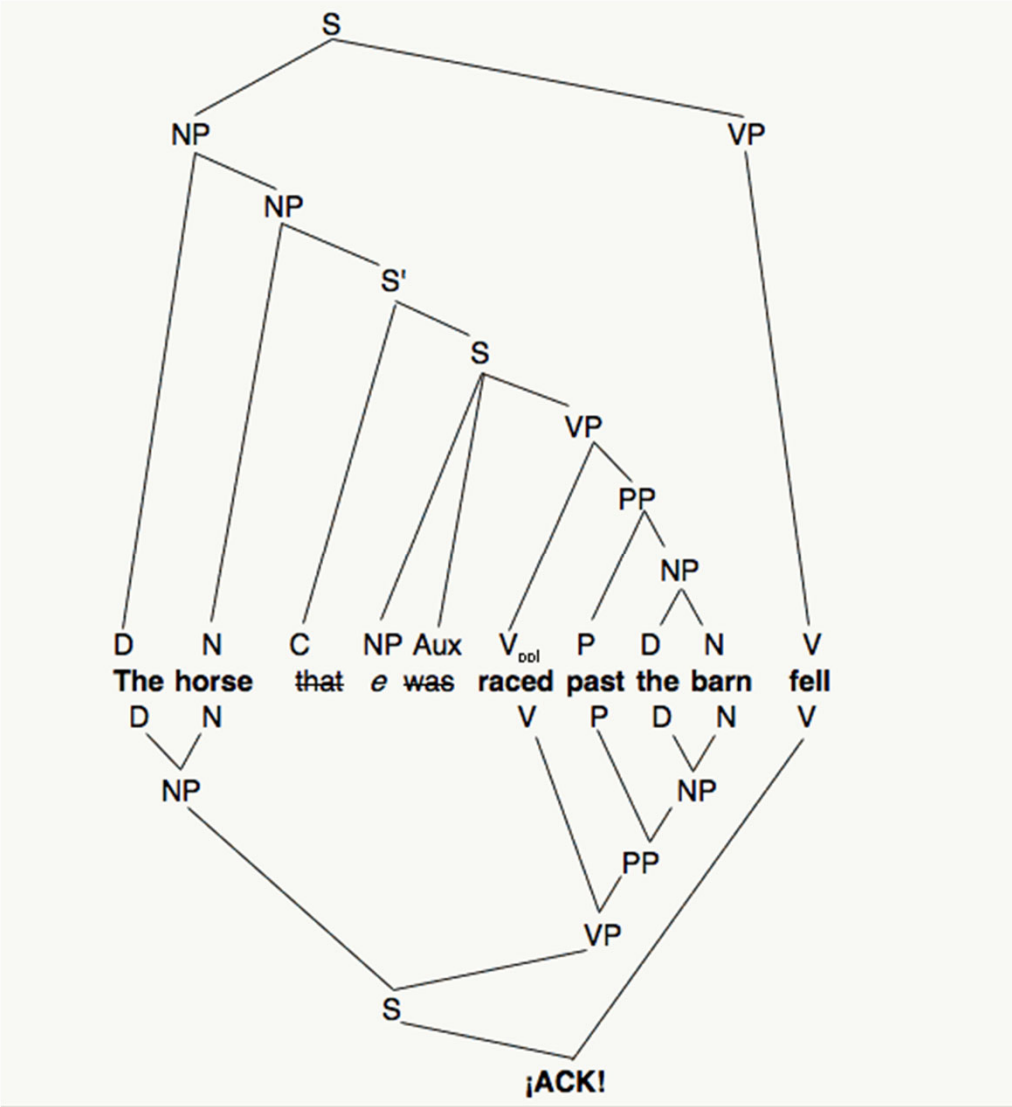
Syntactic explanations of the “garden path” effect

You can see that the grammatical version of this sentence is much more complex syntactically than the failed parse.

Specifically, it has many more nodes.

One strategy you might take to save space is to initially prefer smaller trees.

This is called “minimal attachment”



Syntactic explanations of the “garden path” effect

Another parsing strategy in this framework is “late closure.” The idea here is that there is a bias to keep the current clause “open” as long as possible.

In this case, “with the dynamite” wants to associate with the big verb phrase starting with “blew open.”

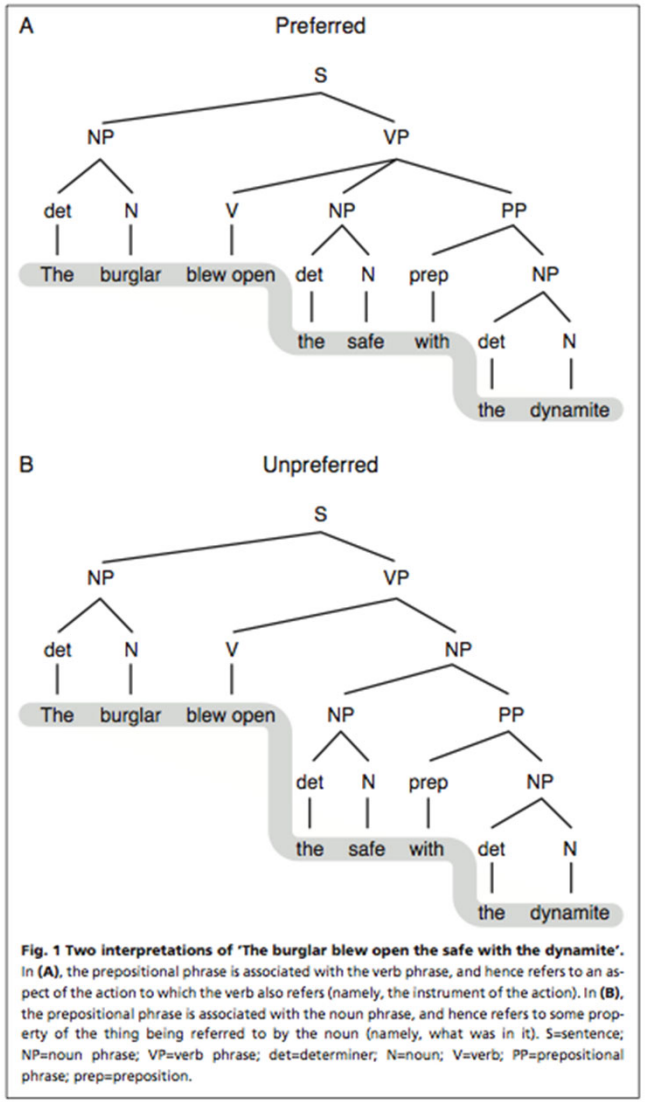


Fig. 1 Two interpretations of ‘The burglar blew open the safe with the dynamite’. In **(A)**, the prepositional phrase is associated with the verb phrase, and hence refers to an aspect of the action to which the verb also refers (namely, the instrument of the action). In **(B)**, the prepositional phrase is associated with the noun phrase, and hence refers to some property of the thing being referred to by the noun (namely, what was in it). S=sentence; NP=noun phrase; VP=verb phrase; det=determiner; N=noun; V=verb; PP=prepositional phrase; prep=preposition.

Frazier, L. (1987) Sentence processing: a tutorial review, in *Attentionand Performance XII: The Psychology of Reading* (Coltheart, M., ed.),pp. 559–586, Erlbaum

Syntactic explanations of the “garden path” effect

We can see late closure at work if we consider a sentence that requires the “unpreferred” parse to be sensible:

“The burglar blew open the safe with the jewels.”

In careful tests of reading time and eye movements, we can see that people are slower to read sentences where the sensible reading requires the unpreferred parse.

That is, you momentarily entertain the idea that the jewels were used to blow open the safe, and then have to retrace your steps.

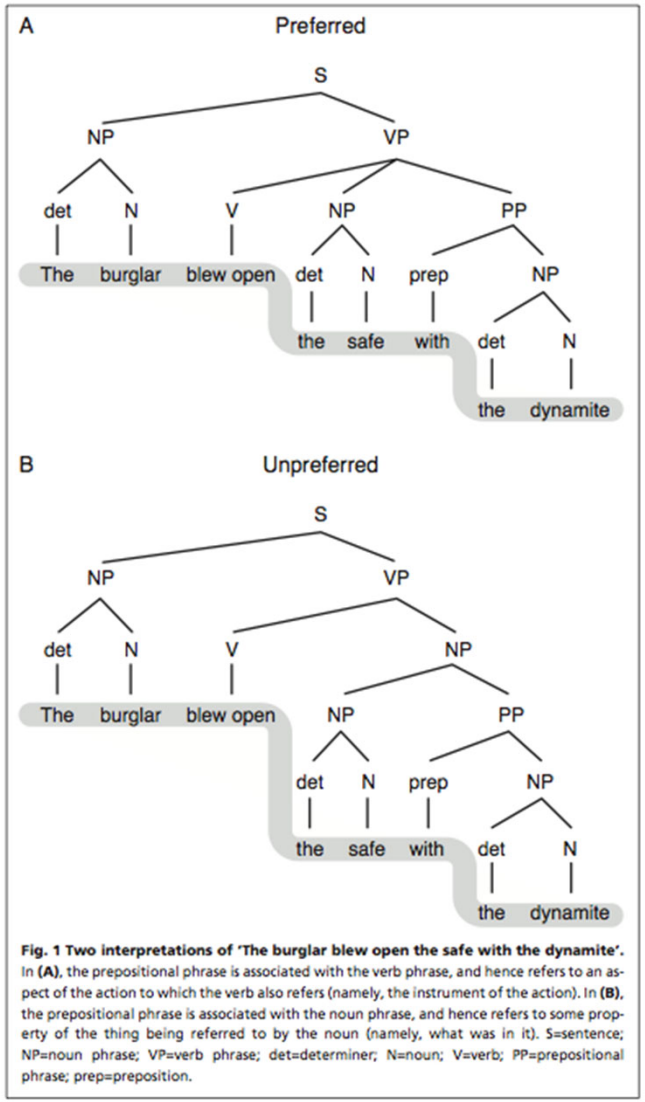


Fig. 1 Two interpretations of ‘The burglar blew open the safe with the dynamite’. In **(A)**, the prepositional phrase is associated with the verb phrase, and hence refers to an aspect of the action to which the verb also refers (namely, the instrument of the action). In **(B)**, the prepositional phrase is associated with the noun phrase, and hence refers to some property of the thing being referred to by the noun (namely, what was in it). S=sentence; NP=noun phrase; VP=verb phrase; det=determiner; N=noun; V=verb; PP=prepositional phrase; prep=preposition.

Syntactic explanations of the “garden path” effect

The explanation rests on the idea of a parser that deals exclusively with the abstract syntactic structure of the sentence (remember the “lumber mill” example from speech perception).

The parser has its own operating characteristics. It wants to make the simplest tree possible (minimal attachment) and it accomplishes this, in part, by leaving clauses open as long as possible (late closure).

This is one way of dealing with ambiguity very *efficiently* – ignore it until it becomes a problem!

“World knowledge” can be very local

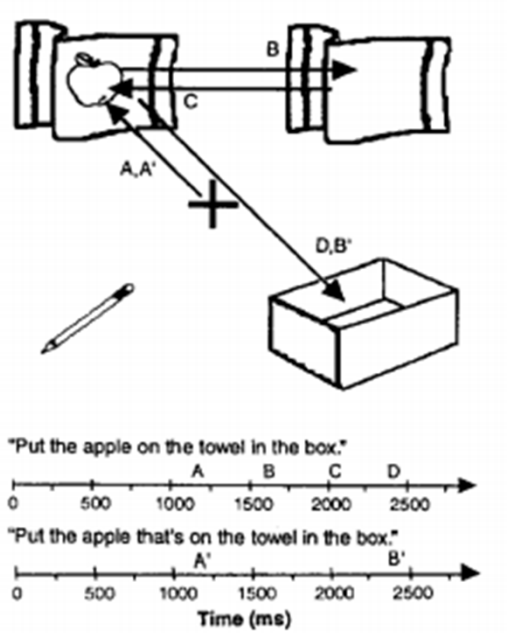


Fig. 1. Typical sequence of eye movements in the one-referent context for the ambiguous and unambiguous instructions. Letters show when in the instruction each eye movement occurred, as determined by the mean latency for that type of eye movement (A' and B' correspond to the unambiguous instruction).

Spivey et al. (1995) gave people a visual world with a pencil, a box, a towel and an *apple on a towel.*

People were then instructed to “put the apple on the towel in the box.”

You can imagine that with minimal attachment, “the towel” would temporarily seem like it should be the end of the clause “put the apple on the towel.”

And lo, people do seem to look at the towel while planning their movements, as though they are considering it as a target, before planning a movement to the box.

“World knowledge” can be very local

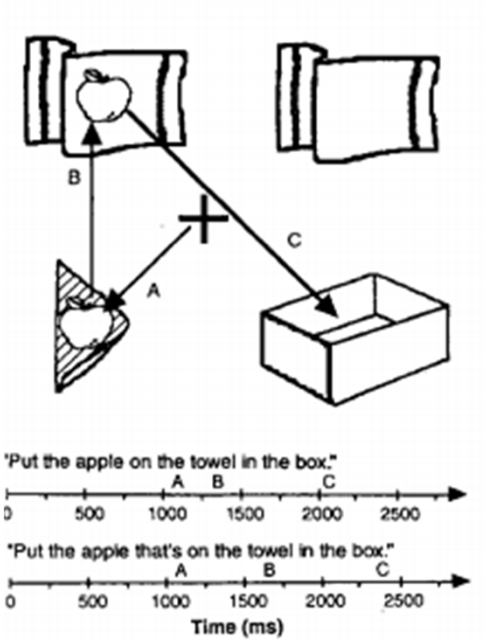


Fig. 2. Typical sequence of eye movements in the two-referent context. Note that the sequence and the timing of eye movements, relative to the nouns in the speech stream, did not differ for the ambiguous and unambiguous instructions.

“World knowledge” can be very local

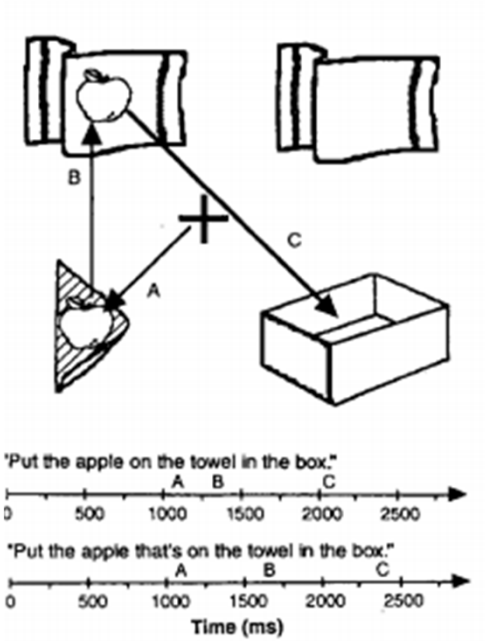


Fig. 2. Typical sequence of eye movements in the two-referent context. Note that the sequence and the timing of eye movements, relative to the nouns in the speech stream, did not differ for the ambiguous and unambiguous instructions.

Now consider what happens when there are *two apples* in the display.

All of a sudden it is *felicitous* to provide more information regarding *which apple you’re talking about.*

So people don’t look at the towel as much.

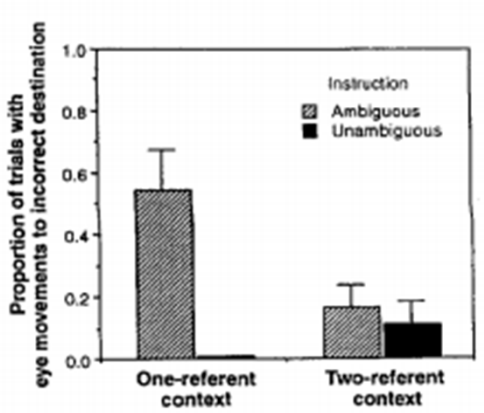


Fig. 3. Proportion of trials in which participants looked at the incorrect destination.

An attractive dichotomy:

Wernicke's aphasia = "Word salad" or "Jabberwocky," often with neologisms. Syntax roughly intact:

> One Wernicke's aphasia patient, for example, replied to a question about his health with: "I felt worse because I can no longer keep in mind from the mind of the minds to keep me from mind and up to the ear which can be to find among ourselves" (Kriendler et al., 1971). Another patient, when asked about his poor vision said: "My wires don't hire right". And an aphasic physician, asked if he was a doctor, replied: "Me? Yes, sir, I'm, a male demaploze on my own. I still know my tubaboys what for I have that's gone hell and some of them go". One aphasic described a fork as "a need for a schedule" and a spoon as "How many schemes on your throat".

Broca's aphasia = "Telegraphic" speech, difficulty with word order and grammatical particles (relative sparing of important content words):

> These fluent but uninterpretable utterances are very different from the Broca aphasic's answer when asked what brought him to the hospital: "Yes – ah – Monday – ah – Dad and P.H. (the patient's name) and Dad ... hospital. Two ... ah doctors ..., and ah ... thirty minutes ... and yes ... ah ... hospital. And, er Wednesday ... nine o'clock and eh Thursday, ten o'clock ... doctors. Two doctors ... and ah ... teeth. Yeah ... fine" (Badecker and Caramazza, 1985).

An attractive dichotomy:

Wernicke's aphasia = problems with activation (inhibition?) in the lexicon.

Broca's aphasia = problems with *grammar*

Note that this is fairly distant from how Broca described his original cases.

Remember this slide from a few weeks ago?

This is confusing: *Broca's actual patients would not necessarily be classified as Broca's aphasics.* They were arguably unable to produce speech at all, apparently due to motor control issues, and would be classified as suffering from *apraxia of speech.*

Paul Broca's presentation of Leborgne (or "Tan")



Behavioral description:

- could only produce one jargon syllable
- seemed otherwise to be in possession of his senses
- in fact, seemed to *understand what was being said.*

Consider:

- the role of dissociations in this argument: why is it critical that Leborgne has other senses intact?

- why does a "capacity for articulate speech" sound like a more plausible function of a particular region of the brain than "criminality" or "philopropogenitiveness?"

- if you could examine Tan, what measurements would you want to make in order to specify his deficits in terms of the levels of description used in modern linguistics (e.g., phonetic, phonological, lexical, syntactic)?

An attractive dichotomy:

Wernicke's aphasia = problems with activation (inhibition?) in the lexicon.

Broca's aphasia = problems with *grammar*

This dichotomy is interesting in part because it seems well aligned with a sharp distinction drawn late 20th Century formal linguistics between the *grammar* and the *lexicon*.

This starts with Chomsky's formulation of the computational problem of language learning, and compelling demonstrations of the independence of syntax and semantics.

It appears that we recognize a new item as a sentence not because it matches some familiar item in any simple way, but because it is generated by the grammar that each individual has somehow and in some form internalized. And we understand a new sentence, in part, because we are somehow capable of determining the process by which this sentence is derived in this grammar.

"Neither (a) 'colorless green ideas sleep furiously' nor (b) 'furiously sleep ideas green colorless', nor any of their parts, has ever occurred in the past linguistic experience of an English speaker. But (a) is grammatical, while (b) is not." (Chomsky, 1955)

Man Bites Dog

Why do we need psycholinguistic tests to observe grammatical difficulties in comprehension?

In conversation, there are lots of redundant cues to what a speaker intends:

The dog bit the man.

There are so many ways to get the right meaning here:

- Most likely combination of DOG, BITE and MAN into an event representation with or without syntax.

- Canonical word order: Agent – Verb – Undergoer; Subject – Verb – Object

Further, a lot of language comprehension takes place in discourse context that clarifies a lot...

Maybe Broca's patients' comprehension depends on these compensatory *heuristics*. If that were the case you might want to test them with sentences more like:

The dog that the man is biting is angry.

Indeed, Caramazza and Zurif devised a list of just this kind of sentence, and a set of control sentences.

TABLE 1

SENTENCE MATERIAL USED IN THE EXPERIMENT

A. Control sentences
1. The girl is kicking a green ball.
2. The cat is chasing a blue bird.
3. The girl is pulling a red wagon.
4. The man is reading a yellow book.
5. The boy is riding a new bicycle.
6. The girl is holding a broken doll.
7. The boy is throwing a round ball.
8. The boy is carrying a sharp pencil.

B. Center-embedded sentences with semantic constraints
9. The apple that the boy is eating is red.
10. The wagon that the horse is pulling is green.
11. The book that the girl is reading is yellow.
12. The bat that the boy is carrying is brown.
13. The bicycle that the boy is holding is broken.
14. The coat that the girl is wearing is new.
15. The dress that the woman is washing is torn.
16. The balloon that the clown is holding is round.

C. Reversible center-embedded sentences
17. The cow that the monkey is scaring is yellow.
18. The horse that the bear is kicking is brown.
19. The cat that the dog is biting is black.
20. The fish that the frog is biting is green.
21. The lion that the tiger is chasing is fat.
22. The girl that the boy is hitting is tall.
23. The man that the woman is hugging is happy.
24. The girl that the boy is tickling is happy.

D. Improbable center-embedded sentences
25. The horse that the girl is kicking is brown.
26. The lion that the baby is scaring is yellow.
27. The bird that the worm is eating is blue.
28. The dog that the man is biting is black.
29. The man that the horse is riding is fat.
30. The woman that the boy is spanking is sad.
31. The policeman that the robber is chasing is skinny.
32. The girl that the cat is patting is small.

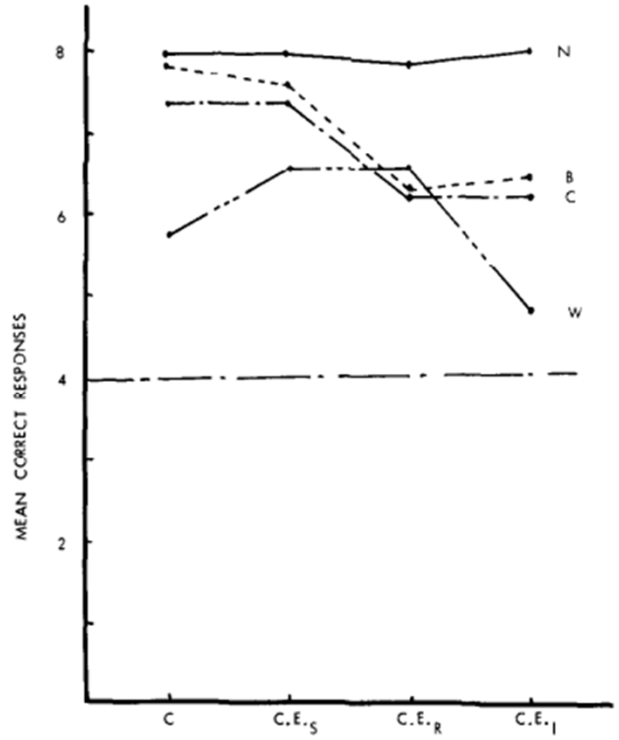


FIG. 1. Mean correct performance for each patient group (N = Normal, B = Broca, C = Conduction, and W = Wernicke) as a function of sentence type (CE_S = semantically constrained, CE_R = reversible, CE_I = improbable, and C = control).

Data from a sentence-picture matching task.

Healthy controls are more or less equally good at all stimulus types.

Wernicke's patients are poor overall. This is probably a result of their comprehension deficits.

The Broca's patients seem to have a particular difficulty keeping "who did what to whom" straight when syntax conflicted with common sense, probable semantic roles for nouns, etc.

This *specific* comprehension deficit was taken as evidence that Broca's aphasia impacts *syntax* specifically.

Problems with this dichotomy

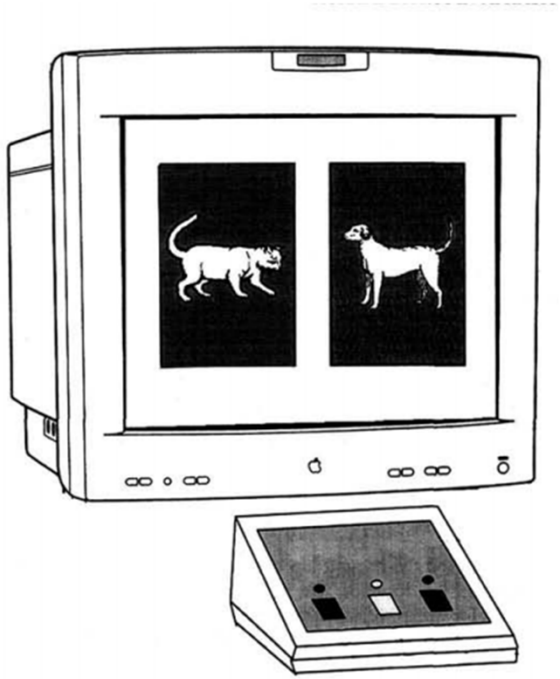


Figure 4. Experimental setup and sample visual stimuli (Experiments 1A and 1B).

Dick et al. had participants listen to four different types of sentences:

Active:

The dog is biting the cat.

Subject cleft:

It's the dog that is biting the cat.

Object cleft:

It's the cat that the dog is biting.

Passive:

The cat is bitten by the dog.

Instructions: Indicate which animal is *doing the action.*

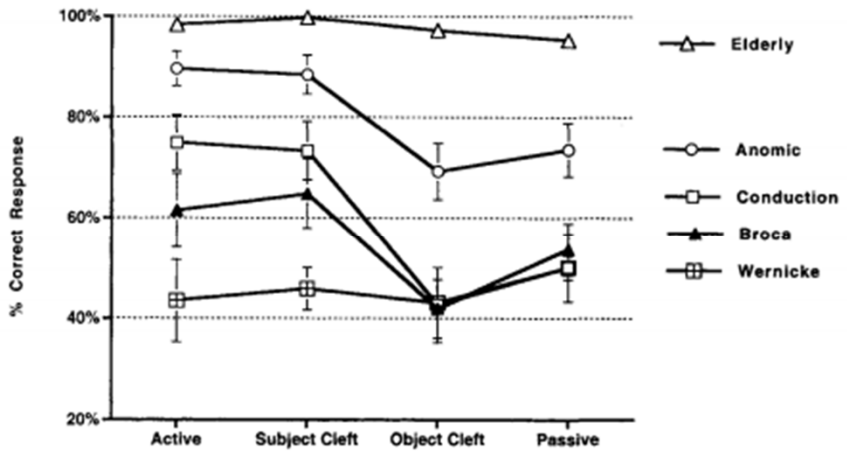


Figure 5. Percentage of correct responses by sentence type for aphasic patients (grouped by the Western Aphasia Battery) and older control participants (Experiment 1A). Error bars represent ±1 *SEM*.

When aphasia patients and healthy aging controls do this task, a pattern emerges much like what Caramazza and Zurif observed. The Broca's aphasia patients have particular difficulty with the syntactically complex stimuli (although note that their performance is overall not so great).

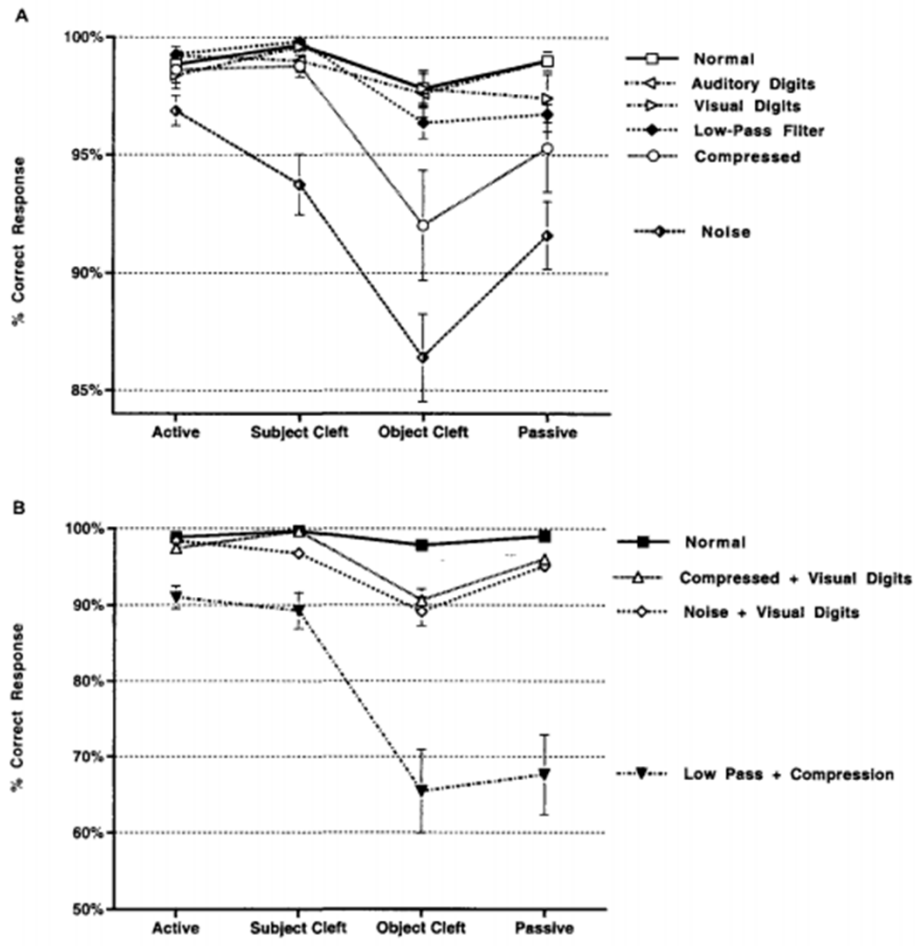


Figure 9. A: Percentage of correct responses by sentence type for students under single-stress conditions (Experiment 1B). B: Percentage of correct responses by sentence type for students under dual-stress conditions (Experiment 1B). Error bars represent ±1 *SEM*.

These are data from healthy undergraduates under conditions that stress the language system in different ways.

Note that the object cleft and passive sentences are particularly vulnerable to all manipulations.

The “stress” conditions were things like speeding up the speech, or asking people to remember lists of numbers as a secondary task.

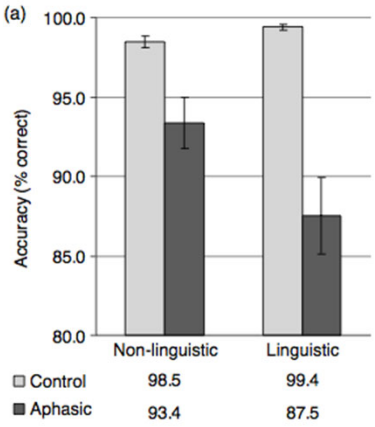
This suggests that the comprehension deficits seen in Broca’s aphasia patients may be a “performance” issue after all.

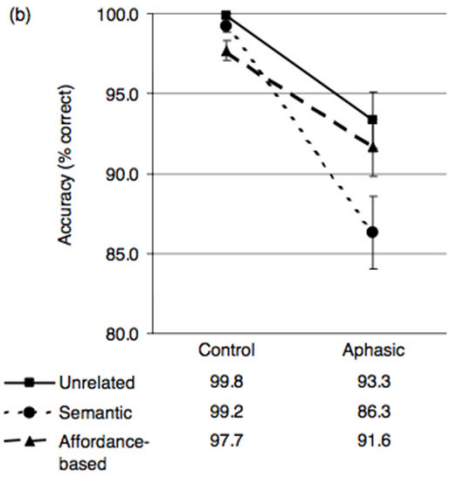
Are Broca's aphasics comprehension deficits language-specific?



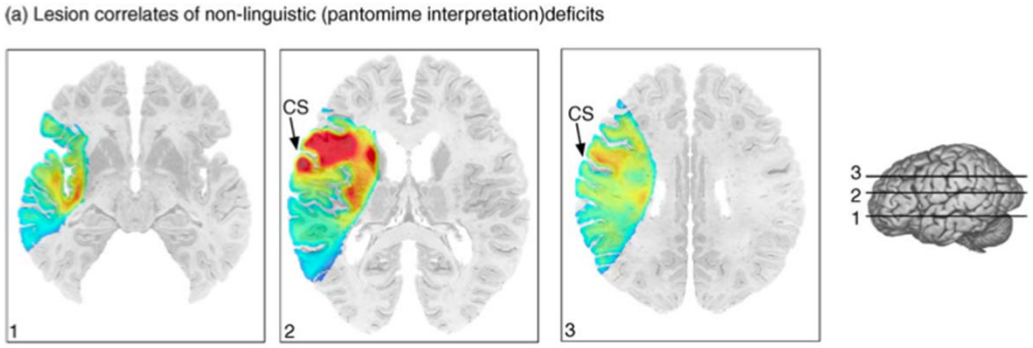
Saygin et al. (2004) asked patients to match the action to the object.

Broca's aphasics had difficulty with this task, whether the stimuli were linguistic or not.





Are Broca's aphasics comprehension deficits language-specific?

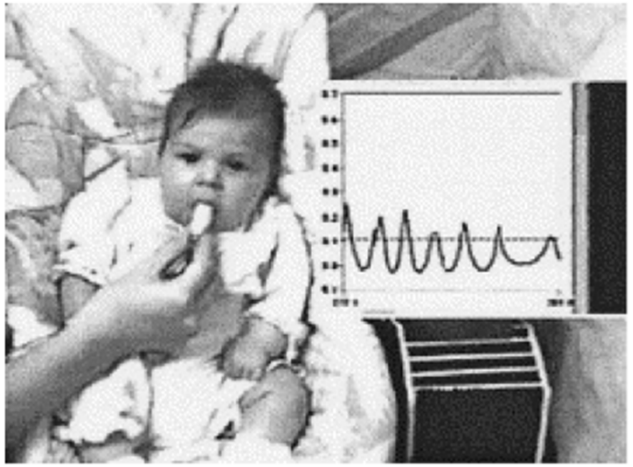


Further, performance in the non-verbal version of this task was significantly correlated with damage to Broca's Area.

Categorical perception in infants



Infant in the "conditioned head turn procedure from Janet Werker's website:
http://infantstudies.psych.ubc.ca/



We'll also look at "high amplitude sucking." (image from an unrelated paper, Rochat & Striano, 2001, IBD)

Categorical perception in infants

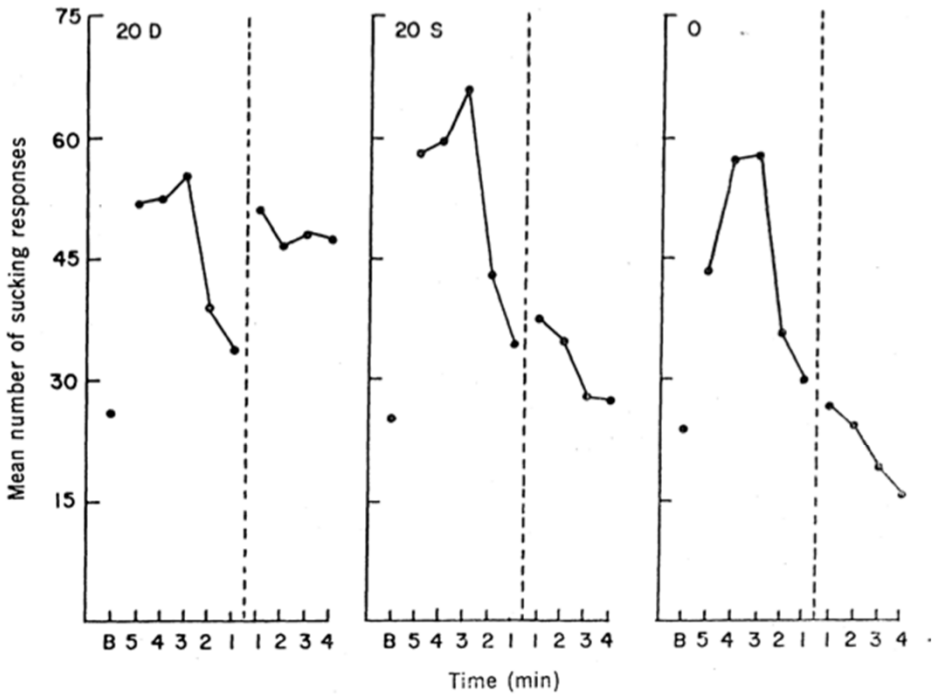
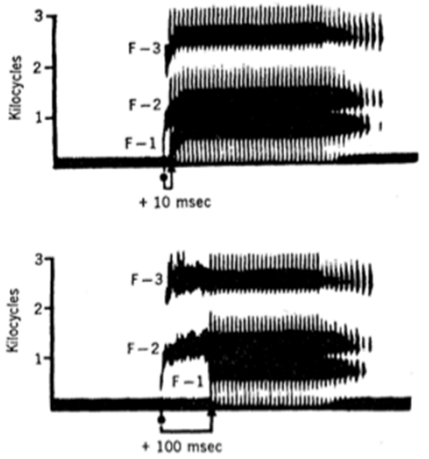


Fig. 2. Mean number of sucking responses for the 4-month-old infants, as a function of time and experimental condition. The dashed line indicates the occurrence of the stimulus shift, or in the case of the control group the time at which the shift would have occurred. The letter *B* stands for the baseline rate. Time is measured with reference to the moment of stimulus shift and indicates the 5 minutes prior to and the 4 minutes after shift.

304

Eimas et al. (1971, Science)



In the main experiment, there were three variations in VOT differences at each of two age levels. In the first condition, 20D, the difference in VOT between the two stimuli to be discriminated was 20 msec and the two stimuli were from different adult phonemic categories. The two stimuli used in condition 20D had VOT values of +20 and +40 msec. In the second condition, 20S, the VOT difference was again 20 msec, but now the two stimuli were from the same phonemic category. In this condition the stimuli had VOT values of −20 and 0 msec or +60 and +80 msec. The third condition, 0, was a control condition in which each sub-

Why was this a big deal?

Categorical perception was considered to be the first step in turning speech into a *symbolic* format that could be used by later linguistic processes.

Four month olds are really little. Some gross motor milestones for that age include:

- manipulating objects with two hands
- holding their head and chest up while prone



Their color vision and general visual acuity are still “in progress.”

They are just starting to produce vowel-like sounds instead of creaky whines.

This is very, very early in development for an apparently language-specific ability to emerge, so it seemed like evidence for innateness.

“Perceptual narrowing” for speech categories

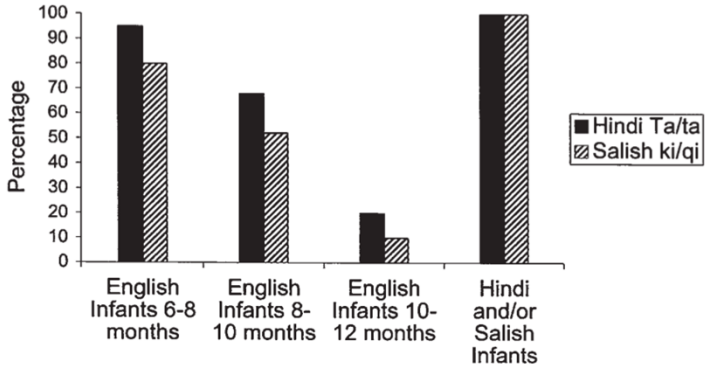


FIGURE 3 The effects of age and experience on infants' ability to discriminate two non-English contrasts. The proportion of infants at each age reaching discrimination criterion on the Hindi and Salish (non-English) contrasts. From "Cross-language speech perception: Evidence for perceptual reorganization during the first year of life," by J. F. Werker & R. C. Tees, 1984a, *Infant Behavior and Development*, 7, 49–63. Adapted with permission of the authors.

At around six months, English-learning infants show evidence of discriminating speech contrasts they've never heard before.

Over the course of development, the ability to perceive these contrasts declines.

How is this related to categorical perception?

A very compelling narrative:

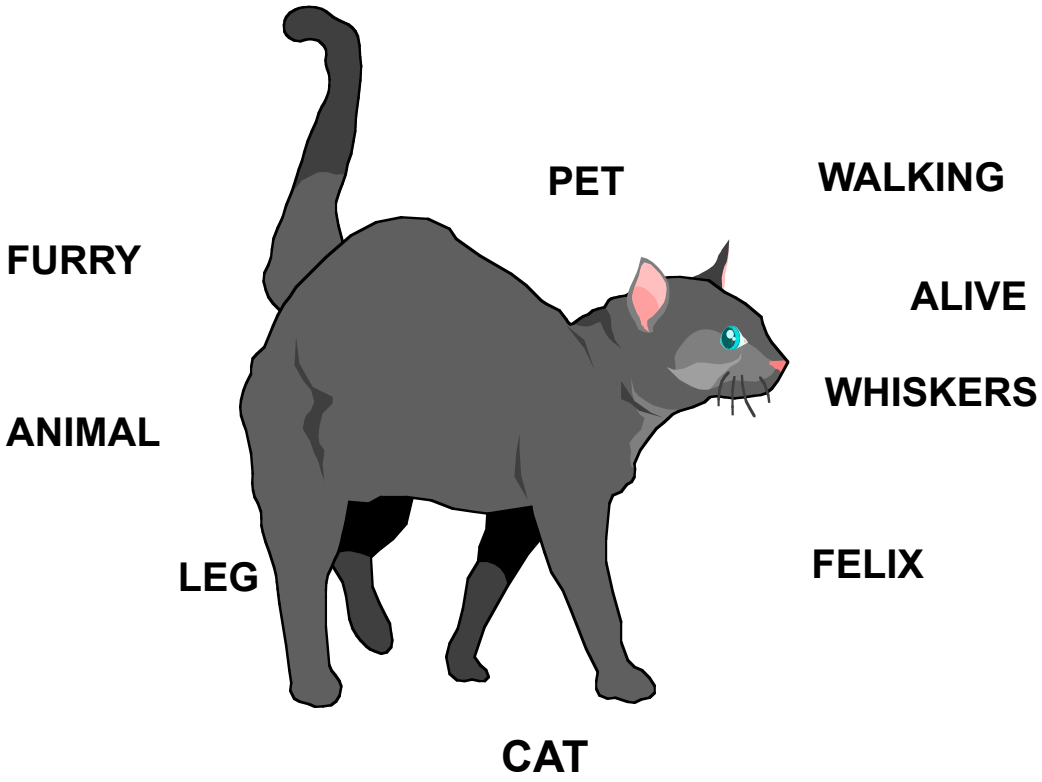
Speech perception depends on categorical perception abilities that are a special property of the language system.

These are present very early in development.

In fact, children start off as "citizens of the world," able to hear all contrasts (i.e., they have *innate, language-general knowledge about speech*) but then rapidly tune their perceptual systems to the specific language in their environment.

The role of *learning*, then, is to *select* among the many possible innately specified phonetic contrasts, and *narrow* down the inventory to just those that are meaningful in the child's native language.

The inductive problem of early word learning

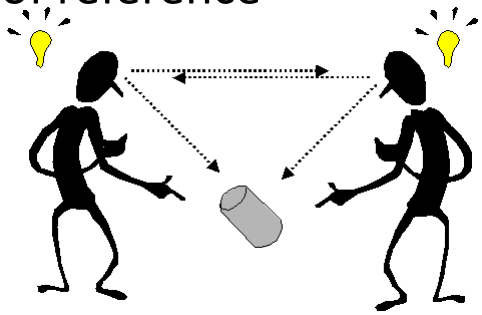


See,e.g, Quine's formulation of the "gavagai" problem:
https://en.wikipedia.org/wiki/Inscrutability_of_reference#Illustration_by_the_use_of_gavagai

How do children acquire word meanings?

Joint attention:

- Sharing an attentional focus with another
- Gaze following emerges around 9 months of age
- Signals an understanding of reference



How do children acquire word meanings?

Joint attention:

- Optimizes word learning
- Baldwin et al.
  - *Joint attention group*: adult interacting with baby labeled the object the baby was playing with
  - *No joint attention group*: voice from behind a screen labeled the object the baby was playing with

Results:

- 18-month-olds learned the novel word in the *Joint attention* group but not in the *No joint attention* group.

How do children acquire word meanings?

How do children narrow the hypothesis space for new words?

Intentional cues from the speaker

e.g., eye gaze direction



Baldwin: 16-17 and 18-19 month olds taught new words

Follow-in labeling: looked at toy that infant was focused on

Discrepant labeling: looked at toy in a bucket

How do children acquire word meanings?

Intentional cues from the speaker

e.g., affective and/or behavioral cues

Akhtar et al.

Infants used adults' facial expression to identify the intended referent of a novel word.

How do children acquire word meanings?

Word learning strategies:

Default strategies that children bring to the word learning situation

Function to narrow the hypothesis space for new words

- e.g., the whole object assumption

How do children acquire word meanings?

e.g., Mutual Exclusivity:(Markman, 1989)

- as a default, objects can have only one name
- Implication: new words map to nameless objects
- Emerges during the second half of the second year





L2 learning as a model system for sensitive period research

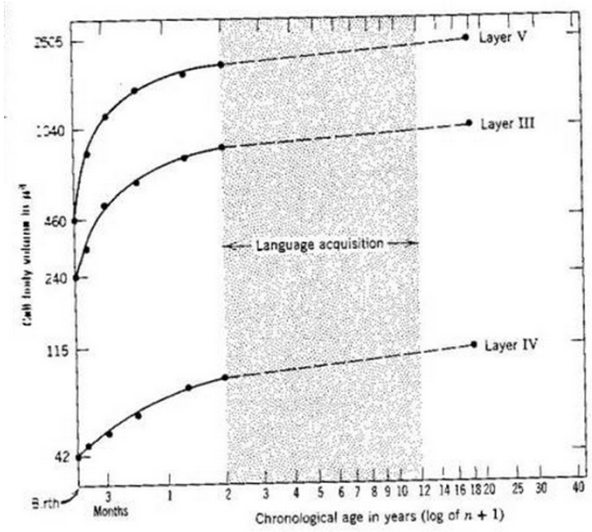


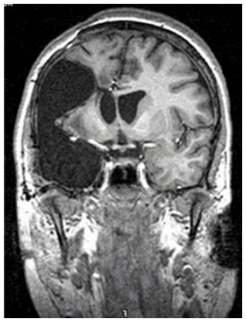
FIG. 4.5. Neurons in the cerebral cortex grow in volume with age. The broken lines between age two and eighteen indicate that no measurements were made on brains of children who died between these ages. (Redrawn from Schadé and van Groenigen, 1961.)

Remember that Lenneberg (1967) developed an influential theory that the sensitive period had to do with the development of cerebral dominance, or developmental changes in language laterality.

This was motivated by

- a rough alignment of the literature on brain development with new insights about language milestones
- a review of the literature on resilience after brain injury or surgery

L2 learning as a model system for sensitive period research





Much of what is known about L1 sensitive periods is from research on idiosyncratic cases, or relatively rare phenomena.

In contrast, many people learn second languages at different times in their lives, often with somewhat limited success. Those limits have also been used to reason about the timing of the sensitive/critical period.

Critical Period Effects in Second Language Learning: The Influence of Maturational State on the Acquisition of English as a Second Language

JACQUELINE S. JOHNSON AND ELISSA L. NEWPORT

University of Illinois

Lenneberg (1967) hypothesized that language could be acquired only within a critical period, extending from early infancy until puberty. In its basic form, the critical period hypothesis need only have consequences for first language acquisition. Nevertheless, it is essential to our understanding of the nature of the hypothesized critical period to determine whether or not it extends as well to second language acquisition. If so, it should be the case that young children are better second language learners than adults and should consequently reach higher levels of final proficiency in the second language. This prediction was tested by comparing the English proficiency attained by 46 native Korean or Chinese speakers who had arrived in the United States between the ages of 3 and 39, and who had lived in the United States between 3 and 26 years by the time of testing. These subjects were tested on a wide variety of structures of English grammar, using a grammaticality judgment task. Both correlational and *t*-test analyses demonstrated a clear and strong advantage for earlier arrivals over the later arrivals. Test performance was linearly related to age of arrival up to puberty; after puberty, performance was low but highly variable and unrelated to age of arrival. This age effect was shown not to be an inadvertent result of differences in amount of experience with English, motivation, self-consciousness, or American identification. The effect also appeared on every grammatical structure tested, although the structures varied markedly in the degree to which they were well mastered by later learners. The results support the conclusion that a critical period for language acquisition extends its effects to second language acquisition. © 1989 Academic Press, Inc.

The data

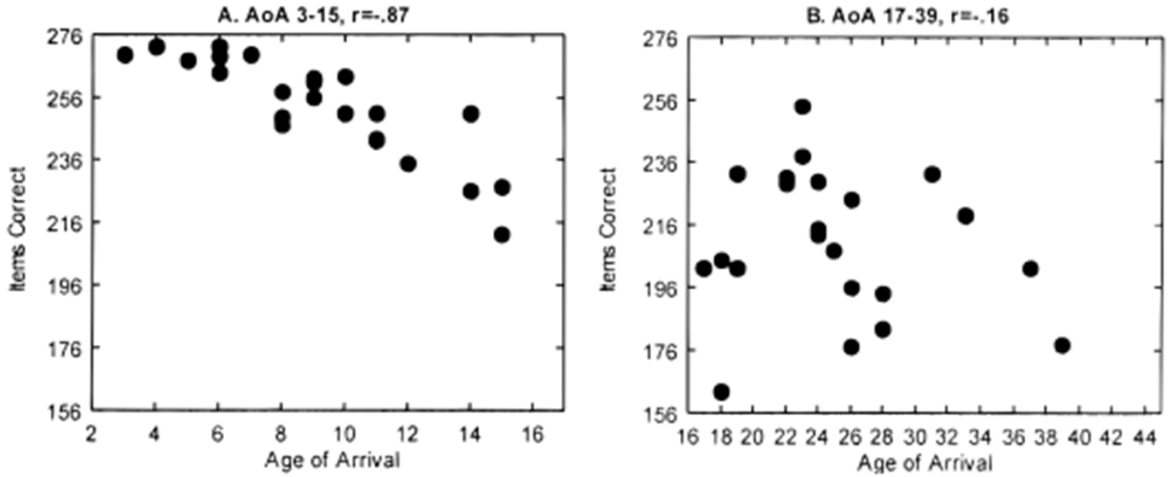


FIG. 1. The age function in J&N89, Early Arrivals (A), and Late Arrivals (B). In this and Figs. 2 and 3, points represent number of items correct for individual participants.

- strong relationship between AOA and accuracy
- early learners (up to about 8) are often near-perfect (native-like)
- nearly random distribution of accuracy for Late Arrivals

Two versions of the “critical period” hypothesis

Version One: The exercise hypothesis. Early in life, humans have a superior capacity for acquiring languages. If the capacity is not exercised during this time, it will disappear or decline with maturation. If the capacity is exercised, however, further language learning abilities will remain intact throughout life.

Version Two: The maturational state hypothesis. Early in life, humans have a superior capacity for acquiring languages. This capacity disappears or declines with maturation.

Which of these do you think JN89 see as more consistent with their data?

Does this change our interpretation of the L1 critical period data?

The “cohort effect.”

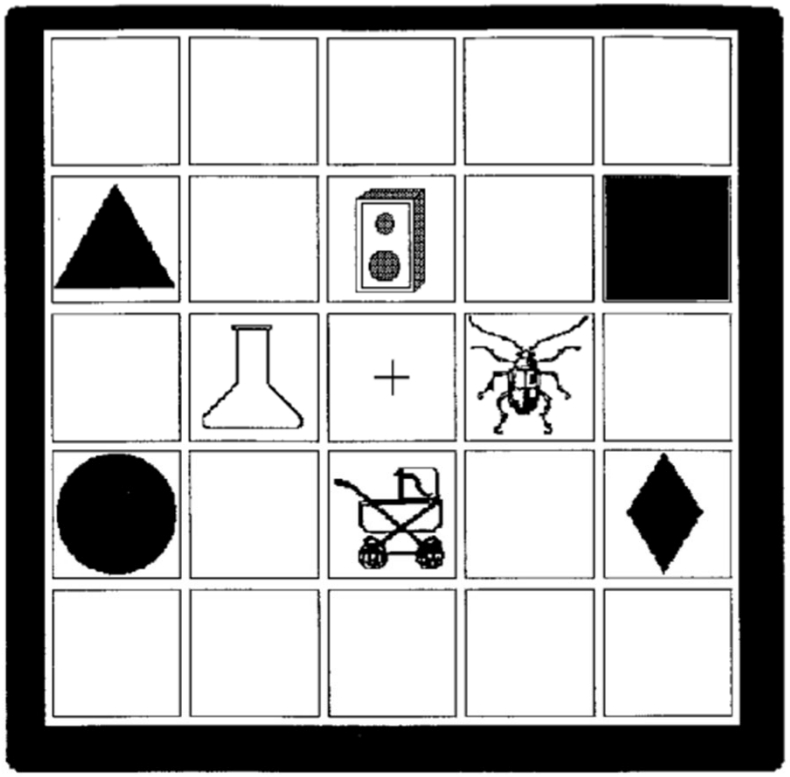


FIG. 3. An example of a stimulus display presented to participants.

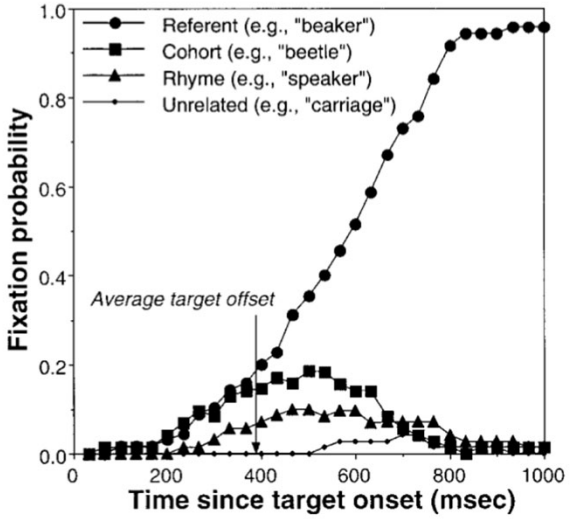


FIG. 4. Probability of fixating each item type over time in the full competitor condition in Experiment 1. The data are averaged over all stimulus sets given in Table 1; the words given in the figure are examples of one set.

Allopenna et al. (1994, JML)

Marian & Spivey examined the cohort effect with "cross-language competitors"

Russian Target			English Competitor	Russian Competitor		
Спички	[spichki]	(Matches)	Spear	Спицы	[spitsi]	(Knitting needles)
Бусы	[boosy]	(Beaded necklace)	Book	Бубен	[buben]	(Tambourine)
Черепаха	[cherepaha]	(Turtle)	Chair	Червяк	[chervyak]	(Worm)
Марка	[marka]	(Stamp)	Marker	Марля	[marlya]	(Cheese cloth)
Баран	[baran]	(Ram)	Barbed Wire	Бархат	[barhat]	(Velvet)
Платье	[platye]	(Dress)	Plum	Плащ	[plashch]	(Raincoat)
Гайка	[gaika]	(Nut)	Gun	Галстук	[galstuk]	(Necktie)
Карта	[karta]	(Map)	Car	Картошка	[kartoshka]	(Potato)
Лак для ногтей	[Lak dlea nogtei]	(Nailpolish)	Lock	Лопата	[lapata]	(Shovel)

English Target	English Competitor	Russian Competitor		
Speaker	Spear	Спички	(spichki)	Matches
Boot	Book	Бубен	(buben)	Tambourine
Shark	Shovel	Шарик	(sharik)	Balloon
Chair	Chess set	Черепаха	(cherepaha)	Turtle
Marker	Marbles	Марка	(marka)	Stamp
Barbed wire	Bark	Бархат	(barhat)	Velvet
Plug	Plum	Платье	(platye)	Dress
Gun	Gum	Гайка	(gaika)	Nut
Lock	Lobster	Лак для ногтей	(Lak dlea nogtei)	Nailpolish

Hypothesis: If L1 words are always activated, native Russian speakers should look at, e.g., a book of matches when the target is "speaker" in English.

Competition during L2 spoken word recognition

Table 2. *Percent of trials in which bilingual participants made eye movements to the competitor items and their corresponding fillers in between-language competition, within-language competition, and simultaneous competition trials in Experiment 1.*

Display	Fixations of between-language competitor	Fixations of within-language competitor
Between-language competitor present	18%	n/a
Within-language competitor present	n/a	18%
Both competitors present	13%	19%
No competitor present; fixations of control filler object	7%	

When only one competitor was present, participants made eye movements to both within- and between-language competitors to the same extent.

When both within- and between-language competitors were present, participants looked at the within-language competitor more, but still looked at the between-language competitor a lot (i.e., twice as often as a control condition).

Competition during **L1** spoken word recognition

Table 4. *Percent of trials in which bilingual participants made eye movements to the competitor items and their corresponding fillers in between-language competition, within-language competition, and simultaneous competition trials in Experiment 2.*

Display	Fixations of between-language competitor	Fixations of within-language competitor
Between-language competitor present	8%	n/a
Within-language competitor present	n/a	14%
Both competitors present	8%	13%
No competitor present; fixations of control filler object	5%	

In contrast, English words do not seem to be activated when (native Russian speakers) do this experiment in Russian.

(The small numerical difference between the between-language and control conditions is not statistically significant.)

A model explaining L1-L2 interactions in terms of spreading activation:
The revised hierarchical model of Kroll & Stewart

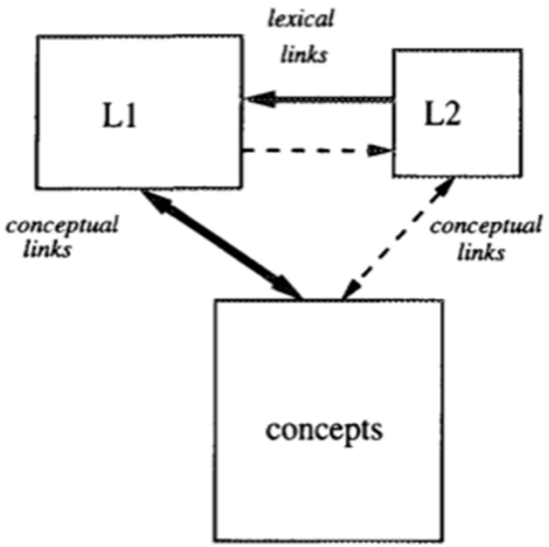


FIG. 3. Revised hierarchical model of lexical and conceptual representation in bilingual memory.

The L1 lexicon is assumed to be:

- larger
- more strongly connected to relevant concepts

The L2 lexicon is assumed to have strong *forward* connections to the L1 lexicon.

That is, activation can spread more readily from an L2 word to an L1 word than the other way.

(This occurs because this direction is emphasized in learning.)

A model explaining L1-L2 interactions in terms of spreading activation:
The Revised Hierarchical Model of Kroll & Stewart (1990; 1994)

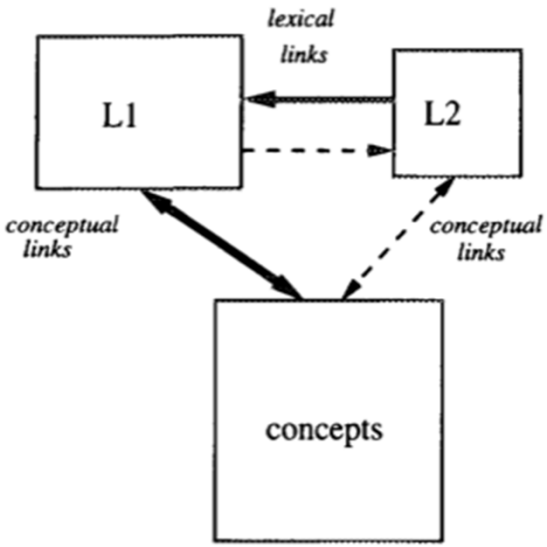


FIG. 3. Revised hierarchical model of lexical and conceptual representation in bilingual memory.

Consider what happens in this model when performing the Marian & Spivey task in L2:

- the concept for “speaker” partially activates the Russian word for “matches”

- this activity feeds back to the conceptual activation for “matches”

- conceptual activation for “matches” leads to eye-movements toward the L1 competitor.

When the task is done in L1, we might imagine that the conceptual links with L2 are too weak to create this kind of interference reliably.

Competition, interference and bilingualism:

Note that this model makes a counter-intuitive prediction even for “balanced bilinguals,” i.e., that they do not look like “two monolinguals in one,” but rather that competitive dynamics and inhibition will be engaged all the time b/c the conceptual links are similarly strong for both languages.

This has lead to a lot of work on bilinguals’ advantages in non-linguistic cognitive control tasks…